

Algorithm for tracking the top two maximums in a single pass:

1- Initialize:

largest = -Infinity

secondlargest = -Infinity

2- Iterate: loop through each number in the array.

Case A (number > largest) if number is greater largest down to secondlargest and largest update.

Case B (number > secondlargest and number != largest) number not equal to largest and greater than 2nd largest update 2nd L with number.

3- Validate: After loop finishes

secondL == -Infinity → [5, 5, 5)

No 2ndL.

Otherwise return 2ndL.

Dry Run:

current number 10

10 > -Infinity

L = 10

2L = -Infinity

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Date: _____

$$5 > 10$$

10 L , 5 ~~1st~~ 2nd L

$$20 > 10$$

20 L , 10 2nd L

$$8 > 20 \times$$

$$8 > 10 \times$$

$$15 > 20 \times$$

$$15 > 10$$

20 L , 15 2nd L

second largest = 15